# ASSIGNMENT 05
# PATIENT APPOINTMENT AUTOMATION SUITE

WORKFLOW OPTIONS USE CASE | Zoho Creator Workflow Configuration and Automation

## 1. Objective

Design an appointment automation layer using all eight workflow options: Form Workflows, Schedules, Approvals, Payments, Blueprints, Batch Workflows, Report Workflows, and Functions.

## 2. Problem Statement

Appointment acknowledgements, reminders, special scheduling approvals, fee updates, and visit summaries can involve repetitive manual work. The automation suite supports timely communication and controlled status changes.

## 3. Scope of the Use Case

- Application: Patient Appointment Booking & Clinic Operations Management
- Base Form/Report: Patient_Appointment_Form and Patient_Appointment_Report
- Users: Patients, Reception Staff, Schedulers, Clinic Manager
- Deliverable: Eight automations covering appointment request through visit follow-up
- Objective: Reduce manual handling, improve reminders, and keep scheduling actions auditable.

## 4. Workflow Options Used and Their Mapping

| Workflow Option | Workflow Name | Purpose | Trigger / Frequency |
|---|---|---|---|
| Form Workflows | On_Appointment_Request_Submit | Acknowledge request and notify reception | Successful form submission |
| Schedules | Appointment_Reminder_Schedule | Send reminders for confirmed visits | Daily at 10:00 AM |
| Approvals | After_Hours_Appointment_Approval | Route after-hours requests to manager | When request is outside clinic hours |
| Payments | Consultation_Fee_Payment_Workflow | Collect configured fee and record payment | After slot confirmation when fee applies |
| Blueprints | Appointment_Lifecycle_Blueprint | Control booking and visit stages | Requested → Review → Confirmed → Checked In → Completed / Cancelled |
| Batch Workflows | Daily_Visit_Reminder_Batch | Send reminders to next-day appointments | Daily at 7:00 AM; batches of 200 |
| Report Workflows | OnEdit_Update_Fee_Estimate | Refresh estimate when visit type changes | Edit of specified fields |
| Functions | Generate_Visit_Summary_Function | Create approved visit summary document | After visit completion or authorized request |

## 5.1 Form Workflows — On_Appointment_Request_Submit

Runs when a new appointment request is added. Sends a receipt with the Appointment ID and explains that staff must confirm the requested slot. Notifies the reception queue.

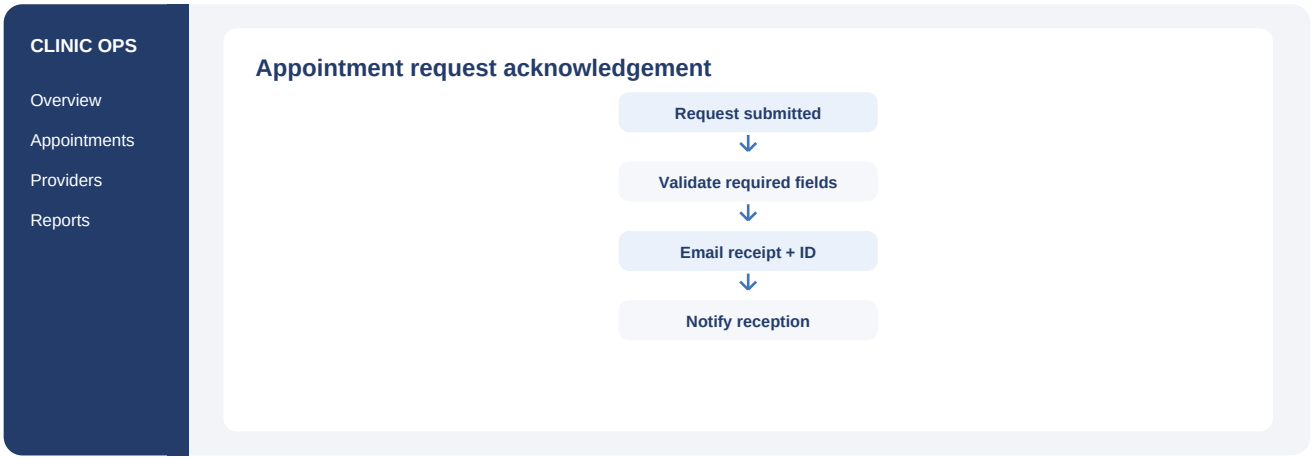



Figure 1: Form workflow acknowledgement and reception notification.

## 5.2 Schedules — Appointment_Reminder_Schedule

Runs daily to identify confirmed appointments that need a reminder. Sends a reminder with clinic, date, time, and rescheduling contact details; it does not send reminders for unconfirmed requests.

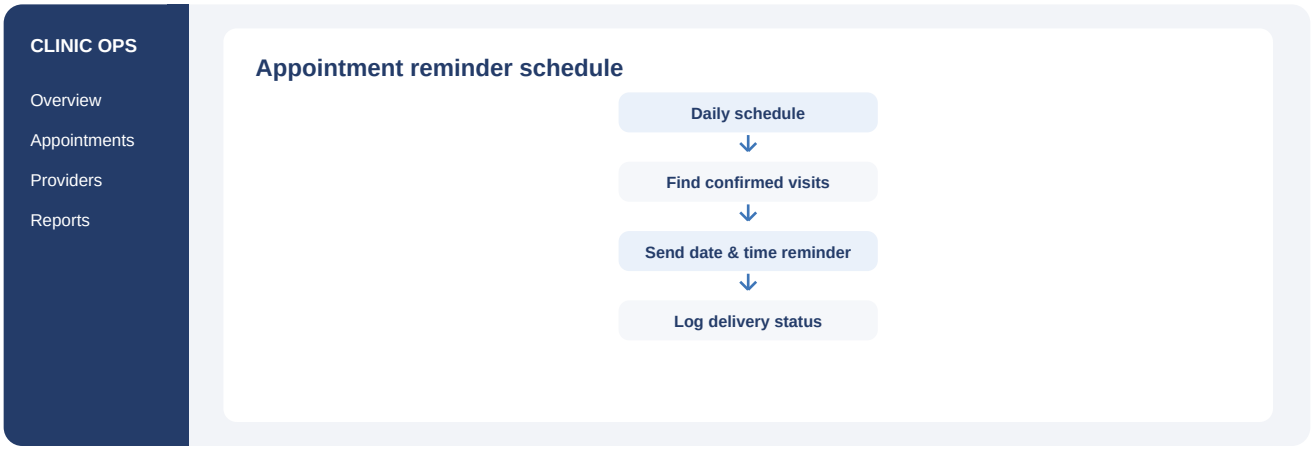



Figure 2: Daily appointment reminder schedule.

## 5.3 Approvals — After_Hours_Appointment_Approval

Routes requests outside standard clinic hours to an authorized manager. The request remains Pending Review until approved or declined; approval does not bypass provider availability checks.

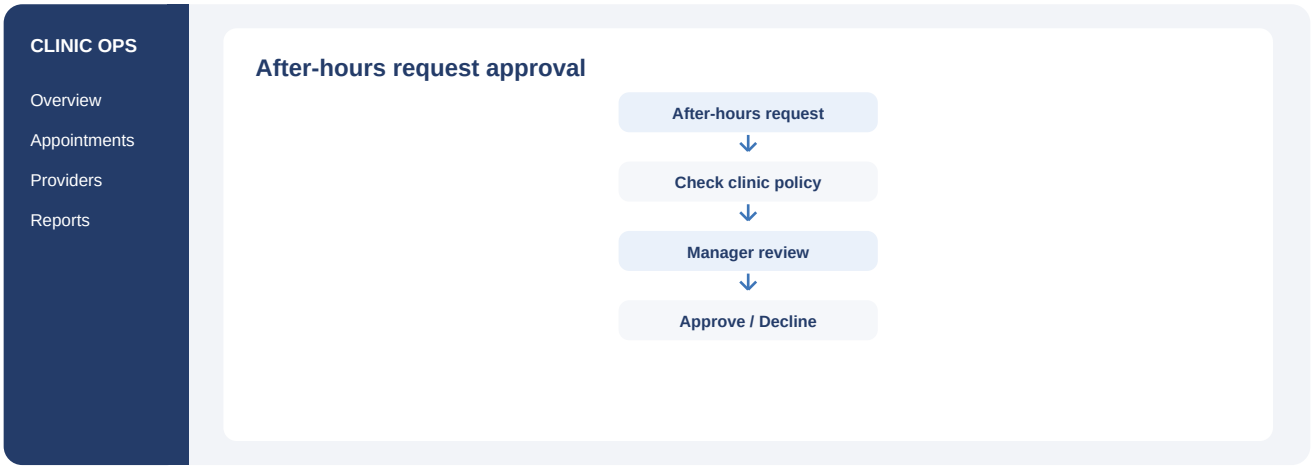



Figure 3: After-hours appointment approval routing.

## 5.4 Payments — Consultation_Fee_Payment_Workflow

For appointments with a configured fee, starts payment after slot confirmation through the connected payment service. A verified success updates Payment_Status and stores the transaction reference. Failure is checked up to three times, then flagged for staff.

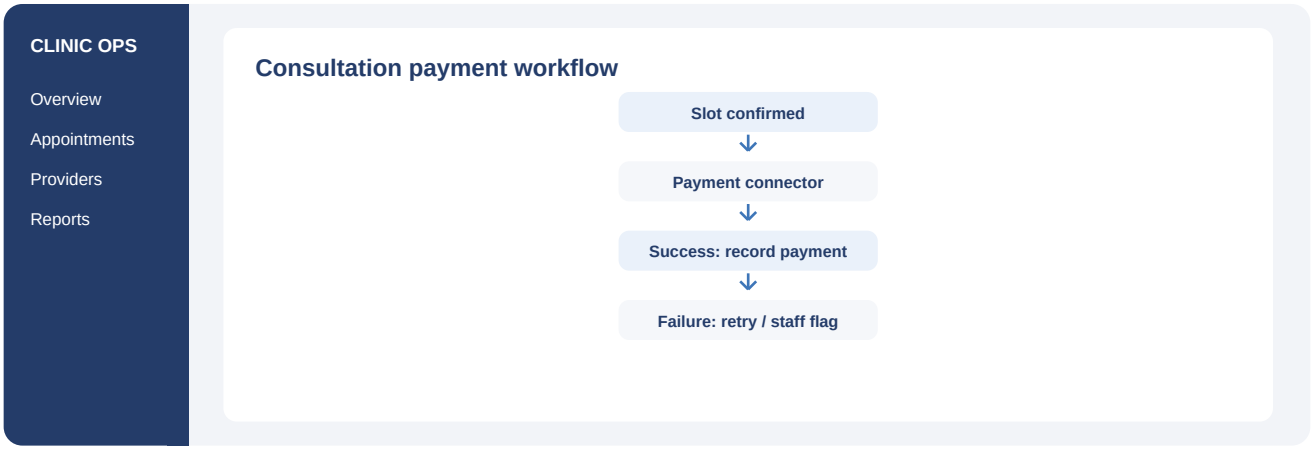



Figure 4: Consultation payment workflow with recovery path.

## 5.5 Blueprints — Appointment_Lifecycle_Blueprint

Uses Requested, Review, Confirmed, Checked In, Completed, and Cancelled stages. Role-based transitions prevent check-in before confirmation and require an audit note for cancellations or overrides.

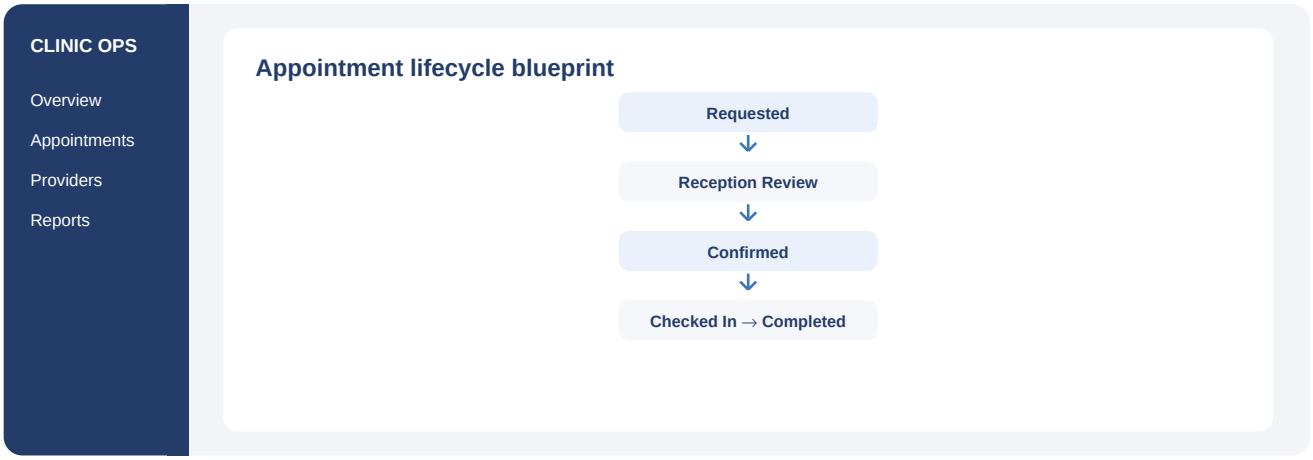



Figure 5: Appointment lifecycle blueprint and transition controls.

## 5.6 Batch Workflows — Daily_Visit_Reminder_Batch

Runs at 7:00 AM to locate confirmed appointments scheduled for the next day. Sends clinic details and arrival instructions in batches of 200 and records delivery status.

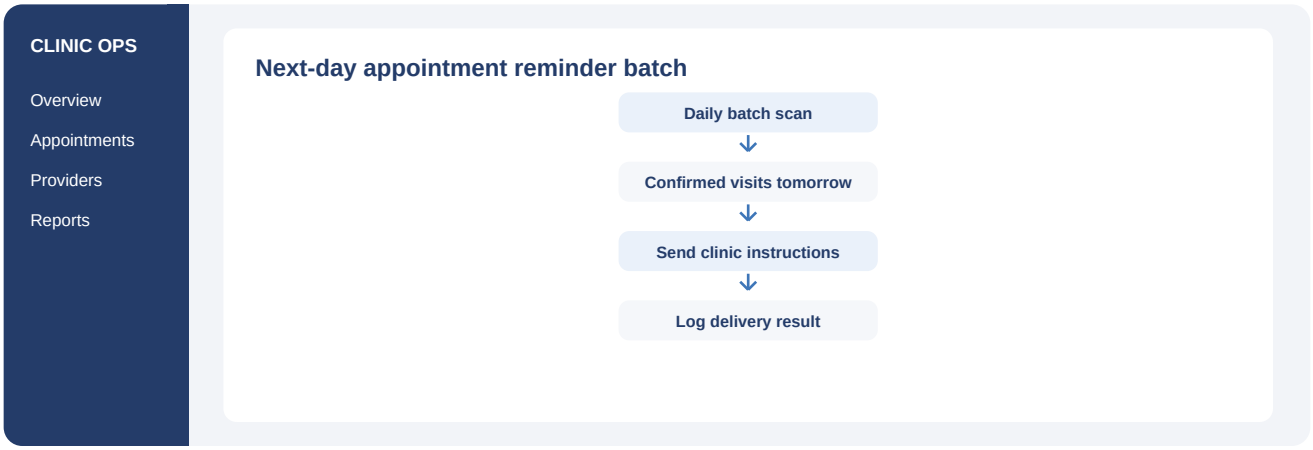



Figure 6: Batch reminder workflow for next-day appointments.

## 5.7 Report Workflows — OnEdit_Update_Fee_Estimate

Triggers when Appointment_Type or Provider_Lookup changes in the appointment report. Refreshes the displayed fee estimate and does not run for unrelated edits. Final charges are confirmed by staff.

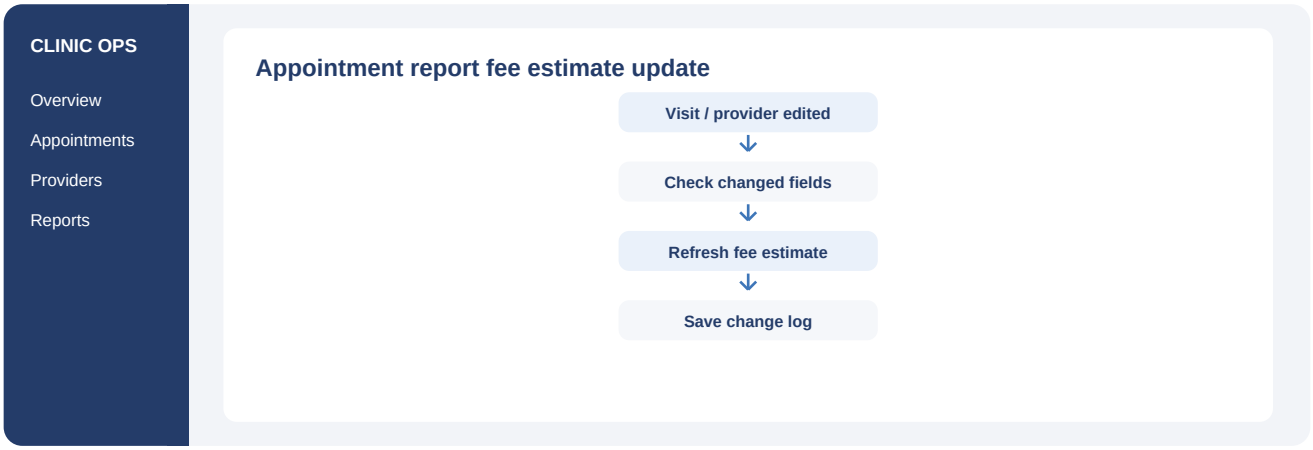



Figure 7: Report edit fee estimate workflow.

## 5.8 Functions — Generate_Visit_Summary_Function

Generates an appointment summary document with appointment reference, provider, date, and administrative follow-up details approved for patient release. It can be called after completion or by an authorized resend action.

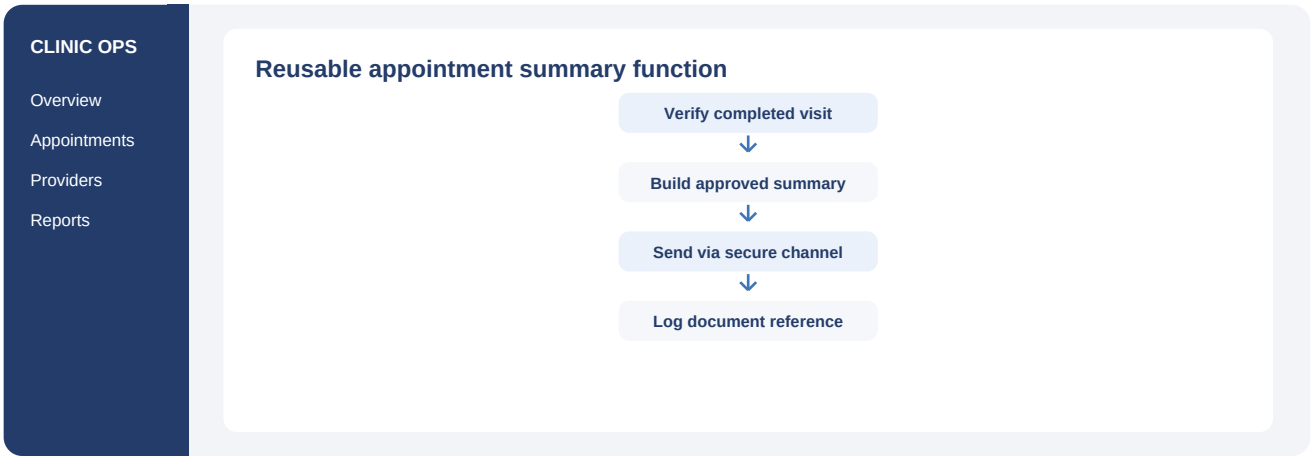



Figure 8: Reusable appointment summary function.

## 6. Configuration and Governance Rules

- Form workflow: log delivery failures without discarding a valid appointment request.
- Schedule: send reminders only for confirmed appointments and retain delivery status.
- Approvals: restrict after-hours decisions to designated managers and keep provider availability checks.
- Payments: use a configured payment connector; verify callbacks up to three times before staff escalation.
- Blueprint: require a reason for cancellation or manual status override; restrict completion to authorized staff.
- Batch workflow: use a batch size of 200 and avoid including sensitive visit reasons in reminders.
- Report workflow: run only when Appointment_Type or Provider_Lookup changes.
- Function: include only approved administrative details and require an authorized user to send the summary.

## 7. Workflow Summary (End-to-End)

- Patient submits Patient_Appointment_Form and receives a request receipt.
- Reception reviews the request and checks provider availability.
- After-hours requests are routed for approval.
- A confirmed appointment triggers payment when a fee applies.
- Blueprint controls confirmation, check-in, completion, and cancellation.
- Daily batch workflow sends reminders for confirmed visits.
- Authorized staff can generate an administrative visit summary after completion.

## 8. Expected Learning Outcomes

- Distinguish event-triggered actions from time-based schedules.
- Use approvals and Blueprints to control appointment transitions.
- Apply batch workflows to reminder delivery.
- Create reusable functions with permission and content checks.
- Apply access controls to patient data and payment actions.

## 9. Deliverable and Result

This report documents the eight workflow options, triggers, governance controls, and appointment lifecycle. Figures are illustrative mockups. Live email, payment, and secure-document services must be configured before automation can run.